RURALEDGEHEALTH

An Offline Mobile Edge AI Reference Architecture for Intelligent IoT Sensor Classification

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Abstract

The rapid growth of the Internet of Things (IoT) has transformed the way health, environmental, and sensor-based systems collect and process data. However, most intelligent IoT applications rely heavily on cloud-based artificial intelligence, continuous internet access, and high-end computing infrastructure. This creates a major challenge in rural and resource-constrained environments, where unstable connectivity, limited power availability, and low-cost mobile devices make cloud-dependent solutions impractical. To address this gap, the present mini-project proposes **RuralEdgeHealth**, an **offline mobile edge AI reference architecture** designed to perform intelligent IoT sensor classification directly on entry-level Android smartphones without depending on remote cloud servers.

RuralEdgeHealth is built as a **modular five-layer architecture** to ensure efficient, secure, and scalable deployment in low-resource settings. The system includes a Presentation and Interface Layer for Bluetooth Low Energy (BLE)-based sensor interaction and manual fallback input, an IoT Sensor Integration Layer for local data acquisition, an Edge Intelligence Layer for on-device machine learning inference, a Data Management and Persistence Layer for encrypted local storage, and an Infrastructure and Runtime Layer for optimized execution on limited hardware. At the core of the system is a lightweight **Random Forest model** converted into **ONNX format**, enabling fast and energy-efficient three-level sensor status classification such as **Normal, Moderate, and Critical**. The architecture is specifically designed for **Android 8.0+ devices with 2 GB RAM**, making it suitable for deployment on low-cost smartphones commonly available in underserved communities.

The system demonstrates strong technical feasibility, achieving **98.25% classification accuracy**, **sub-100 ms inference latency**, an **active memory footprint below 200 MB**, and **battery consumption under 4.0% per hour** during mixed-use operation. These results show that offline mobile edge AI can provide a practical and reliable alternative to cloud-based IoT intelligence in areas with limited digital infrastructure. By enabling local inference, secure data handling, and low-cost deployment, RuralEdgeHealth contributes to the broader goal of inclusive and accessible smart sensing technologies for rural healthcare and other real-world edge computing applications.

Modifications and Enhancements to Improve the Project

Although RuralEdgeHealth already offers a strong foundation for offline IoT intelligence, several improvements can make the project **more efficient, effective, reliable, and deployment-ready**. **INT8 post-training quantization** can reduce the model size from approximately **3.5 MB to under 1 MB**, lowering CPU overhead and improving inference speed on low-end devices. The BLE sensing module can be enhanced with **adaptive sampling and battery-aware optimization**, allowing the system to reduce polling frequency during repeated normal classifications and skip non-essential computations when battery level is low. The architecture can also be extended with **localized privacy-preserving personalization**, enabling the model to gradually adapt to a user's own baseline sensor patterns without external cloud retraining.

Further reliability can be achieved through **real-world field testing, sensor calibration, anomaly detection, and confidence-based prediction support**, helping the system handle noisy or inconsistent sensor readings more accurately. In addition, **peer-to-peer alert relaying** through Wi-Fi Direct or BLE advertising can improve emergency communication in areas without internet connectivity. User-centered enhancements such as **multilingual support, voice-guided interaction, simplified manual input forms, and a more accessible interface for non-technical users** can significantly improve practical usability in rural communities. Together, these modifications would strengthen RuralEdgeHealth as a scalable, human-centered, and sustainable mobile edge AI solution for intelligent IoT deployment in underserved environments.

Keywords: Edge AI, Internet of Things (IoT), Mobile Edge Computing, On-Device Inference, Offline AI, BLE, Random Forest, Quantization, Resource-Constrained Environments